Dinesh Chandra Gaddam

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Machine Learning Engineer focused on developing LLM-powered systems, agentic AI workflows, and end-to-end machine learning pipelines. Experienced in retrieval-augmented generation (RAG), LLaMA-based models, knowledge graph integration, and reinforcement learning, with a strong foundation in data analysis, modelling and evaluation with scalable AI solutions.

SKILLS

Agentic AI & LLM Systems: Agentic Workflows, Retrieval-Augmented Generation (RAG), Tool-Augmented LLMs, ReAct-style Reasoning, Context Management, LLM Orchestration, Multi-step Reasoning Pipelines
ML Engineering & Systems: End-to-End ML Systems, Model Deployment, REST APIs (FastAPI), ETL Pipelines, Data Engineering, Batch Processing, System Integration, Evaluation Pipelines, Model Monitoring
Programming & Data: Python (PyTorch, TensorFlow, scikit-learn, Pandas, NumPy), SQL, PySpark
Cloud & Infrastructure: AWS (S3, SageMaker), Docker, CI/CD, API Integrations, Cloud-native design.
Visualization & Analytics: Tableau, Power BI, Looker

PROFESSIONAL EXPERIENCE

Associate, Premier Life Sciences LLC (Contract) **August – October 2026**

- Analyzed 13,000+ units produced monthly using Python and pandas, tracking an average fault rate of 0.5% and identifying quality trends and recurring defects.
- Investigated a 5% parts scrap rate using time-series analysis trend and seasonality analysis and hypothesis testing; developed Power BI reports to communicate production trends and quality deviations to stakeholders.

Research Engineer, Betterway **May - July 2026**

- Designed and deployed an end-to-end MLOps inference and monitoring pipeline for healthcare risk scoring, processing **59,753** patient stays with zero dropped requests during live model rollbacks.
- Achieving a production test PR-AUC of 0.222 ($1.9\times$ baseline lift) and 0.670 ROC-AUC while maintaining strict flagged-share stability within a 18%–24% operational band.

Teaching Assistant, GWU Data Science Department **September – December 2025**

- Guided around 20 students on **ML system design, evaluation pipelines, error analysis** emphasizing production-ready thinking.
- Supported implementation and debugging of **end-to-end ML workflows** including fine tuning, deployment and integration challenges.

AI/ML Research Assistant, GWU Data Science Department **June – December 2025**

- Built a **production-oriented LLM pipeline** using LLaMA-3 to extract and structure FDA drug-label data, enabling automated ingestion for 10000 drugs into downstream reasoning, analytics, AI systems.
- Engineered knowledge graph via NetworkX from drug data to support schema-aligned integration and analysis.
- Designed **schema-aligned data integration workflows** combining heterogeneous datasets (PrimeKG, Hetionet), improving link prediction performance (AUC: 0.894, F1: 0.81) for the 10M+ edges.

Research Data Scientist, Bharat Smart Services Company(startup) **June 2024 – Mar 2025**

- Developed an appliance detection system for energy disaggregation by building an LSTM-based classifier using feature engineering and model tuning, improving accuracy from 86% to 92%.
- Designed ETL pipelines to ingest, process household energy data enabling analysis across 1,000+ households.
- Built and trained a CNN-based appliance image detection model on **AWS SageMaker** with data stored in **S3**, achieving 98% F1; containerized the inference service with Docker **and deployed** via Render with automated GitHub Actions CI/CD pipeline.

TECHNICAL EXPERIENCE

Health-Aware-School-Meal-Recommendation-system, Reinforcement Learning

- Developed a contextual bandit-based meal recommendation system using data from 170 Fairfax County schools to balance user preferences with nutritional requirements; **reduced regret by 76% compared to baseline methods.**

RAG-Based Agentic AI System. Vector DB, Langchain

- Designed and deployed an agentic AI system using LangChain to enable multi-step reasoning over unstructured data, implementing with FAISS-based retrieval, context management, and dynamic query routing.
- Built **an LLM orchestration layer combining retrieval, summarization, and response generation** with **ReAct-style reasoning,** improving response relevance through context-aware retrieval.

EDUCATION

M.S, **Data Science**, The George Washington University — January 2024 – December 2025
B. Tech, **Electronics and Communication Engineering**, NIT, India — August 2019 – May 2023

PUBLICATIONS & CERTIFICATIONS

Published “Temperature Forecasting on the Jena Climate Dataset Using SARIMA and Box-Jenkins Models" at FTC Conference.
Review "Health-Aware Meal Recommendation System" at EAAI